

Milestone 2

Group 3

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1. Final Feature Set and Data Description

The dataset consists of 7,043 customer records and 21 variables. After cleaning, 7,032 observations remained. Eleven records were removed due to non-numeric entries in the TotalCharges variable.

The target variable is *Churn*, a binary variable indicating whether a customer discontinued the service (1 = Yes, 0 = No). The distribution shows that approximately 26.5% of customers churned, indicating moderate class imbalance.

The dataset includes 19 descriptive features after removing the customerID identifier. These consist of:

- **Numerical features:**
 - tenure (0–72 months)
 - MonthlyCharges (~18–120)
 - TotalCharges (0–9000+)
- **Categorical features:**
 - Demographic attributes (gender, SeniorCitizen, Partner, Dependents)
 - Service-related variables (InternetService, OnlineSecurity, TechSupport, etc.)
 - Account information (Contract, PaymentMethod, PaperlessBilling)

Data quality inspection revealed that TotalCharges was incorrectly stored as an object type. It was converted to numeric format and invalid entries were removed. No additional missing values were observed.

The dataset was split into training (80%) and testing (20%) subsets using stratified sampling to preserve class proportions.

2. Model Implementation

Two predictive models were implemented to compare performance: Logistic Regression and Gradient Boosted Decision Trees (GBDT). Both models were built using a preprocessing pipeline to ensure consistent handling of numerical and categorical features.

Numerical features (tenure, MonthlyCharges, TotalCharges, SeniorCitizen) were standardized using StandardScaler. Categorical features were encoded using OneHotEncoder with handle_unknown="ignore". The preprocessing and classification steps were combined using a scikit-learn Pipeline.

Logistic Regression

Logistic Regression was used as a baseline model due to its interpretability and strong performance on structured data. The following configuration was used:

- Regularization: L2
- $C = 1.0$
- Maximum iterations = 1000

Gradient Boosted Decision Trees

Gradient Boosting was selected as a more advanced information-based method capable of capturing non-linear relationships. The configuration used was:

- Number of estimators = 200
- Learning rate = 0.05
- Maximum tree depth = 3
- Random state = 42

Both models were trained on 80% of the data using stratified sampling and evaluated on the remaining 20% test set.

3. Model Evaluation

To evaluate model performance, the following metrics were used: accuracy, precision, recall, F1-score, and ROC-AUC. Since the dataset exhibits class imbalance, recall and F1-score are particularly important to assess the model's ability to correctly identify customers who are likely to churn.

The results are summarized below:

Metric	Logistic Regression	Gradient Boosting
Accuracy	0.804	0.792
Precision	0.648	0.631
Recall	0.572	0.529
F1-score	0.608	0.576

ROC-AUC	0.836	0.839
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4. Model Selection

Although Gradient Boosting achieved a slightly higher ROC-AUC, Logistic Regression demonstrated superior recall and F1-score. Since recall is critical in churn prediction to minimize missed churners, Logistic Regression is selected as the preferred model for deployment.

Additionally, Logistic Regression offers higher interpretability, making it easier to explain predictions and risk factors to business stakeholders.